

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 2

VRAESTEL

Kopiereg voorbehou

Module EBN111

Elektrisiteit en Elektronika

Mei 2018

Department of Electrical, Electronic and
Computer Engineering

Semester Test 2

QUESTION PAPER

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Module EBN111

Electricity and Electronics

May 2018



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		15 + 1 Answer sheets	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

STEPS TO FOLLOW

Do step 0 before the start of the session.

Do step 1 in the first 90 minutes of the test.

Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min):

- Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test.
- Write down the answers for each question on the PRACTICE ANSWER SHEET at the end of the QUESTION PAPER. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111S2**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- Do not use commas! Please make use of full stops.

- Complex numbers:

Rectangular form: always place the “j” **at the start** of the imaginary part of the complex number.

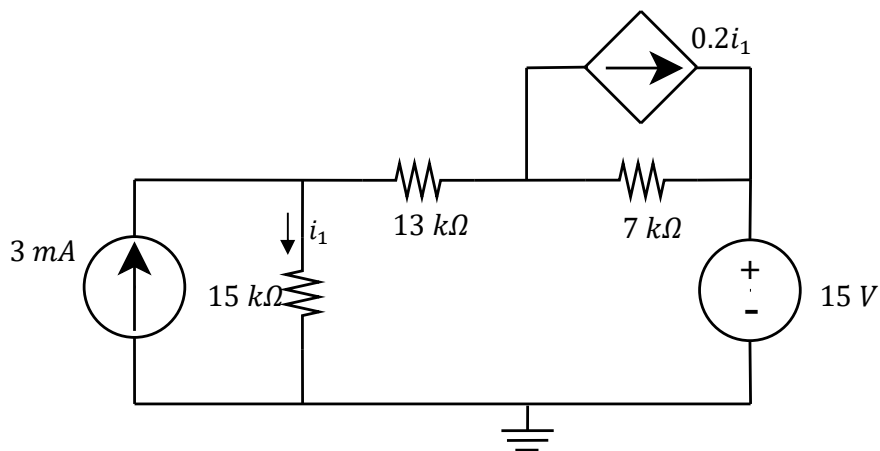
Polar form: always add the “°” (degree sign) after the angle, and make a clear “∠” (angle sign).

Examples

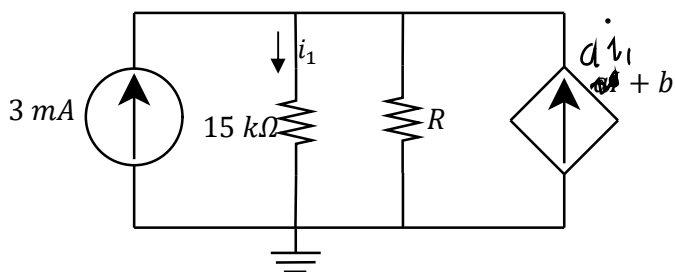
01	–	1	5	4	.	9				k	Ω						
02	1	.	2							r	a	d	÷	s			
03	–	1	.	4	+	j	8			μ	A						
04	1	4	0	°													
05	3	4	0	∠	4	5	°			V							
06	–	j	5	4						k	Ω						
07	1	∠	2	0	°												
08	5	0	∠	–	1	3	0	°		m	A						

SECTION / AFDELING 1 [12]

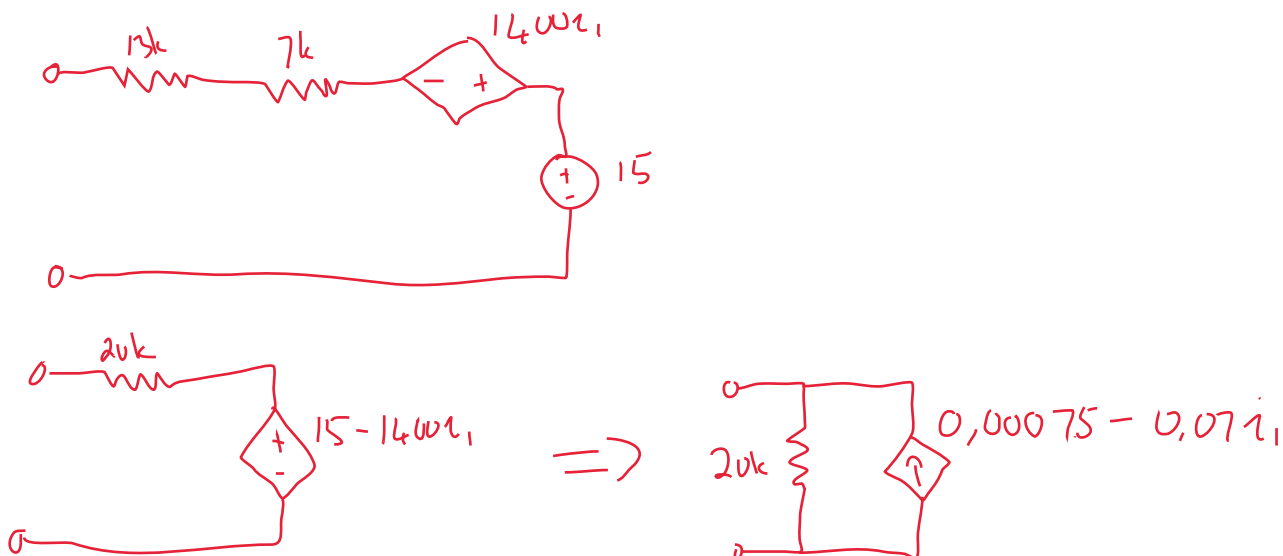
Questions 1-3 [4]		Vraag 1-3 [4]	
Simplify the circuit (a) with source transformation so that it looks like circuit (b) and determine:		Vereenvoudig kringbaan (a) m.b.v. brontransformasie sodat dit lyk soos kringbaan (b), en bepaal:	
01	R	01	R
02	a	02	a
03	b	03	b



(a)



(b)



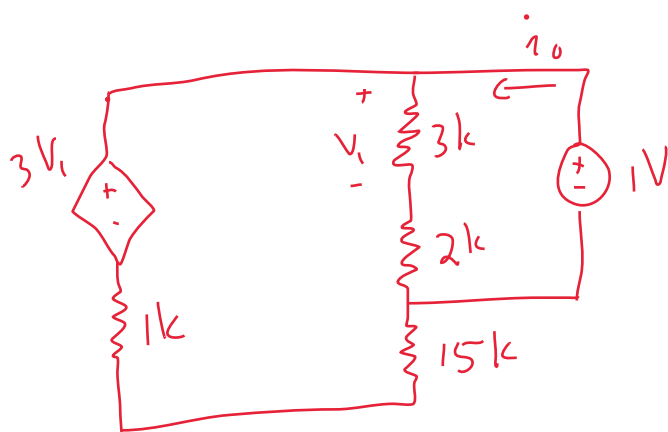
Questions 4-5 [4]		Vraag 4-5 [4]	
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals $a-b$.		Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} met verwysing tot terminale a en b .	
04	V_{TH}	04	V_{TH}
05	R_{TH}	05	R_{TH}

$$21k i - 3V_1 - 15 = 0 \quad \text{but } V_1 = 3k i$$

$$12k i = 15$$

$$i = 1,25 \text{ mA}$$

$$V_{TH} = 3k(1,25\text{mA}) + 2k(1,25\text{mA}) = \underline{6,25V}$$



$$V_1 = \frac{3}{5}(1) = 0,6V$$

$$i_0 = \frac{1 - 3(0,6)}{16k} + \frac{1}{5k}$$

$$= 0,15 \text{ mA}$$

$$R_N = \frac{1}{i_0} = 666,67 \Omega$$

Question 6-7 [4]		Vraag 6-7 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals a - b .		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale a en b .	
06	I_N	06	I_N
07	R_N	07	R_N

Handwritten notes in the diagram:
 $12k\Omega$
 $6k\Omega$
 $15V$
 I_1
 I_2
 I_N
 $6k\Omega$
 $12k\Omega$
 $12k\Omega // 6k\Omega = 4k\Omega$
 $6k\Omega // 12k\Omega = 4k\Omega$

$$I_N = I_1 - I_2 = \frac{7,5}{12k} - \frac{7,5}{6k}$$

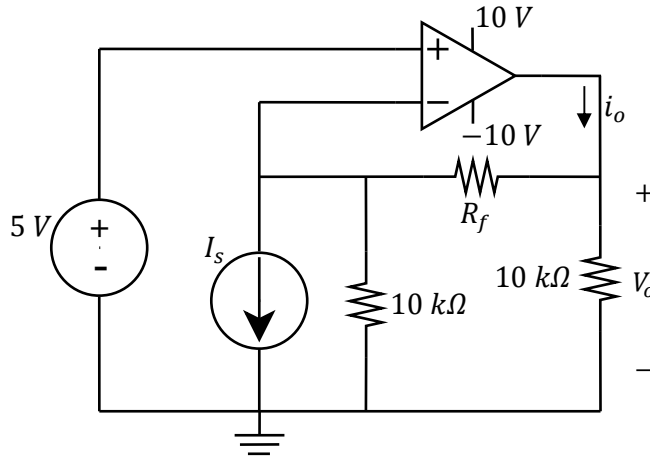
$$= -0,625mA \rightarrow$$

$$R_N = 12k\Omega // 6k\Omega + 6k\Omega // 12k\Omega$$

$$= 8k\Omega \rightarrow$$

SECTION/AFDELING 2 [16]

Question 8-9 [4]		Vraag 8-9 [4]	
08	What is the maximum value of R_f before the operational amplifier reaches saturation, given $I_s = 0\text{ A}$?	08	Gegee $I_s = 0\text{ A}$, wat is die maksimum waarde van R_f voordat die operasionele versterker versadiging bereik?
09	Calculate V_o if $I_s = 2\text{ mA}$ and $R_f = 1\text{ k}\Omega$.	09	Bereik V_o indien $I_s = 2\text{ mA}$ en $R_f = 1\text{ k}\Omega$.



$$8) -10 \leq V_o \leq 10$$

$$-10 \leq \left(1 + \frac{R_f}{10k}\right)(5) \leq 10$$

$$-30k \leq R_f \leq 10k$$

$$\dots \underline{R_{f, \max} = 10k\Omega} \rightarrow$$

$$9) \text{ KCL: } 2m + \frac{5}{10k} + \frac{5 - V_o}{1k} = 0$$

$$\times 10k: 20 + 5 + 50 - 10V_o = 0$$

$$\underline{V_o = 7,5V} \rightarrow$$

Questions 10-11 [4]		Vraag 10-11 [4]	
10	Find v_s if $I_1 = 0$ A and $v_1 = 12$ V.	10	Bepaal v_s indien $I_1 = 0$ A en $v_1 = 12$ V.
11	Find v_1 if $I_1 = 5$ mA and $v_s = 0$ V.	11	Bepaal v_1 indien $I_1 = 5$ mA en $v_s = 0$ V.

$$10) \quad v_1 = - \frac{15k}{10k} (-v_s) \Rightarrow \underline{v_s = 8V} \quad (\text{inverting})$$

$$11) \quad v_a = (5m)(5k) = 25V$$

$$v_1 = \left(1 + \frac{15k}{10k}\right)(25) \quad (\text{non-inverting})$$

$$= \underline{62.5V}$$

Question 12-13 [4]		Vraag 12-13 [4]	
12	Find v_2 if $v_3 = 5\text{ V}$ and $v_1 = 0\text{ V}$.	12	Bepaal v_2 indien $v_3 = 5\text{ V}$ en $v_1 = 0\text{ V}$.
13	Find v_3 if $v_2 = 0\text{ V}$ and $v_1 = 9\text{ V}$.	13	Bepaal v_3 indien $v_2 = 0\text{ V}$ en $v_1 = 9\text{ V}$.

$$\begin{aligned}
 12) \quad 5 &= -10k \left(\frac{12}{5k} + \frac{v_2}{2k} - \frac{v_2}{1k} \right) \quad (\text{summing}) \\
 -\frac{29}{10k} &= -\frac{v_2}{2k} \Rightarrow \underline{v_2 = 5,8\text{ V}}
 \end{aligned}$$

$$13) \quad \frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (\text{difference})$$

$$\begin{aligned}
 \therefore v_3 &= \frac{10k}{5k} (9 - 12) \\
 &= -6\text{ V}
 \end{aligned}$$

Questions 14-15 [4]		Vraag 14-15 [4]	
Find R_1 and R_2 if $V_0 = -13\text{ V}$ and $V_X = 18\text{ V}$.		Bepaal R_1 en R_2 indien $V_0 = -13\text{ V}$ en $V_X = 18\text{ V}$.	
14	R_1	14	R_1
15	R_2	15	R_2

$$14) \cdot 18 = \left(1 + \frac{40k}{R_1}\right)(12) \quad (\text{non-inverting})$$

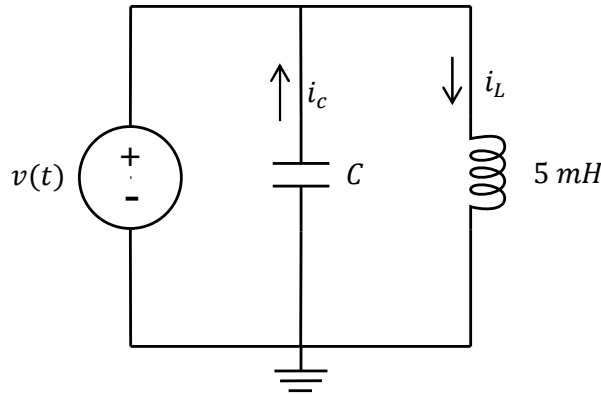
$$\underline{R_1 = 80k\Omega} \rightarrow$$

$$15) -13 = -R_2 \left(\frac{12}{15k} + \frac{18}{10k} \right) \quad (\text{summing})$$

$$\underline{R_2 = 5k\Omega} \rightarrow$$

SECTION/AFDELING 3 [10]

Question 16-17 [4]		Vraag 16-17 [4]	
16	Find $v(1ms)$ if $i_L(t) = 5 + 3e^{-2t} mA$	16	Bepaal $v(1ms)$ indien $i_L(t) = 5 + 3e^{-2t} mA$.
17	Find C if $i_C(10s) = -5 A$ and $v(t) = 5t^2 - 20t V$.	17	Bepaal C indien $i_C(10s) = -5 A$ en $v(t) = 5t^2 - 20t V$.



$$\begin{aligned}
 16) \quad v(t) &= 5m \frac{d}{dt} (5 + 3e^{-2t})m \\
 v(1m) &= (5m)(-6m) e^{-2(1m)} \\
 &= -29,94 \mu V \rightarrow
 \end{aligned}$$

$$\begin{aligned}
 17) \quad i_c(t) &= -C \frac{d}{dt} (5t^2 - 20t) \\
 i_c(10) &= -C (10(10) - 20) \\
 C &= \frac{5}{80} \\
 &= 62,5 mF \rightarrow
 \end{aligned}$$

Question 18-19 [4]		Vraag 18-19 [4]	
Assume steady-state conditions (direct-current) for the circuit below.		Aanvaar gestadigde toestand (gelykstroom) vir onderstaande kringbaan.	
18	Find V_S if the energy stored in the capacitor is 150 mJ.	18	Bepaal V_S indien die energie gestoor in die kapasitor gelyk is aan 150 mJ.
19	Find V_I if the energy stored in the inductor is 2 μ J.	19	Bepaal V_I indien die energie gestoor in die induktor gelyk is aan 2 μ J.

$$18) 150 \text{ m} = \frac{1}{2} (120 \mu) V_C^2 \Rightarrow V_C = \pm 50 \text{ V}$$

$$\therefore \underline{V_S = \pm 100 \text{ V}}$$

$$19) 2 \mu = \frac{1}{2} (10 \text{ m}) I_L^2 \Rightarrow I_L = \pm 0,02 \text{ A}$$

$$V_I = (5 \text{ k}) (\pm 0,02) \\ = \underline{\pm 100 \text{ V}}$$

Question 20 [2]		Vraag 20 [2]	
Determine the equivalent capacitance between terminals a-b if $C_1 = 120\mu\text{F}$ and $C_2 = 150\mu\text{F}$.		Bepaal die ekwivalente kapasitansie tussen terminale a-b indien $C_1 = 120\mu\text{F}$ en $C_2 = 150\mu\text{F}$	
20	C_{ab}	20	C_{ab}

$$\begin{aligned}
 C_{ab} &= (C_1 // C_1 + C_1 + C_1) // C_2 \\
 &= 100 \mu\text{F}
 \end{aligned}$$

→

SECTION / AFDELING 3 [8]

Question 21-22 [4]		Vraag 21-22 [4]	
21	Simplify the following expression of $\vec{V}$ and write the answer in polar form : $\vec{V} = \frac{(\sqrt{8} + j\sqrt{8})^2}{\sqrt[3]{-j8} + 4.685 \angle 92.07^\circ}$	21	Vereenvoudig die volgende uitdrukking vir $\vec{V}$ en skryf die antwoord in polêre vorm : $\vec{V} = \frac{(\sqrt{8} + j\sqrt{8})^2}{\sqrt[3]{-j8} + 4.685 \angle 92.07^\circ}$
22	The following sinusoid is given: $v(t) = 20 \cos(\omega t + 60^\circ) \text{ mV}$ Find $v(10\text{ms})$ if the frequency is $f = 50 \text{ Hz}$.	22	Die volgende sinusgolf word gegee: $v(t) = 20 \cos(\omega t + 60^\circ) \text{ mV}$ Bepaal $v(10\text{ms})$ indien die frekwensie $f = 50 \text{ Hz}$ is.

$$\begin{aligned}
 21) \quad \vec{V} &= \frac{(4 \angle 45^\circ)^2}{\sqrt[3]{8 \angle -90^\circ} + 4.685 \angle 92.07^\circ} \\
 &= \frac{16 \angle 90^\circ}{2 \angle -30^\circ + 4.685 \angle 92.07^\circ} \\
 &= 4 \angle 23^\circ \rightarrow
 \end{aligned}$$

$$\begin{aligned}
 22) \quad v(10\text{ms}) &= 20 \cos\left(2\pi(50)(10\text{ms}) + \frac{60\pi}{180}\right) \text{ m} \\
 &= -10 \text{ mV} \rightarrow
 \end{aligned}$$

Questions 23-25 [4]		Vraag 23-25 [4]	
Given two sinusoid signals v_1 and v_2 where $v_1 = 2 \cos(10t - 123^\circ)$, $v_2 = -2.015 \sin(10t - 23.04^\circ)$.		Twee sinusgolwe v_1 en v_2 word gegee: $v_1 = 2 \cos(10t - 123^\circ)$, $v_2 = -2.015 \sin(10t - 23.04^\circ)$.	
23	Determine $v = v_1 - v_2$ and write the final answer in polar form .	23	Bepaal $v = v_1 - v_2$ en skryf die finale antwoord in polêre vorm .
24	Which statement is correct (write either 1 or 2)? 1. v_1 lags v_2 2. v_1 leads v_2	24	Watter stelling is korrek (skryf of 1 of 2)? 1. v_1 loop na v_2 2. v_1 loop voor v_2
25	What is the phase angle ϕ_d between v_1 and v_2 ? (Note: $0 \leq \phi_d \leq 180^\circ$)	25	Wat is die fashoek ϕ_d tussen v_1 en v_2 ? (Let op: $0 \leq \phi_d \leq 180^\circ$)

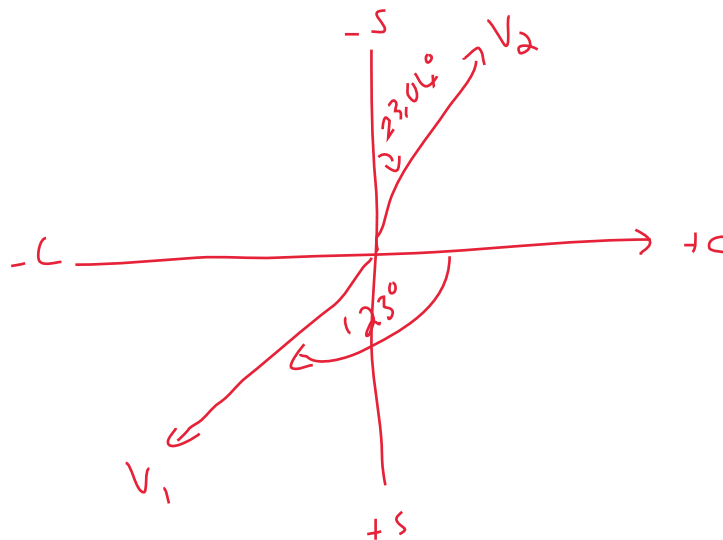
$$23) \quad V_2 = 2.015 \cos(10t - 23.04^\circ + 90^\circ)$$

$$= 2.015 \cos(10t + 66.96^\circ)$$

$$V = 2 \angle -123^\circ - 2.015 \angle 66.96^\circ$$

$$= 4 \angle -118^\circ$$

24)



v_1 leads v_2

$$25) \quad \phi_d = (180^\circ - 123^\circ) + 90^\circ + 23.04^\circ$$

$$= 170.04^\circ$$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	S	2		
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Answer

Unit

01	$R = 20$	$k \Omega$
02	$a = -0,07$	(No unit)
03	$b = 0,00075$	(No unit)
04	$V_{TH} = 6,25$	V
05	$R_{TH} = 6666,67$	Ω
06	$I_N = -0,625$	mA
07	$R_N = 8$	$k \Omega$
08	$R_f = 10$	$k \Omega$
09	$V_o = 7,5$	V
10	$v_s = 8$	V
11	$v_l = 62,5$	V
12	$v_2 = 5,8$	V
13	$v_3 = -6$	V
14	$R_l = 80$	$k \Omega$
15	$R_2 = 5$	$k \Omega$
16	$v(1ms) = -29,94$	mV
17	$C = 62,5$	mF
18	$V_s = \pm 100$	V
19	$V_l = \pm 100$	V
20	$C_{ab} = 100$	nF
21	$\vec{v} = 4 \angle 23^\circ$	
22	$v(10ms) = -10$	mV
23	$\vec{v} = 4 \angle -118^\circ$	
24	Correct statement = 2	(No unit)
25	$\phi_d = 170,04$	°